School of Computing & Mathematical Sciences

Computing and Information Systems

Undergraduate Project Handbook

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1. Introduction

The project is a final part of the undergraduate programme. It is worth 60 credits (half of the credits of your final year) and it is a major undertaking which can require up to 600 hours of work to complete.

The Final Year Projects administrator is:

George Samakovitis
e-mail: G.Samakovitis@gre.ac.uk
phone: 020 8331 9853

Programme Leader and Project Coordinators for the following degrees:

Darrenlloyd Gent
Tel: +44 (0)20 8331 8513
Room: QM360
e-mail: D.T.Gent@greenwich.ac.uk
Games, Digital Media and Computing with Games Development Projects

BSc (Hons) Digital Media Technologies
BSc (Hons) Web Technologies
BSc (Hons) Games Design & Development
BSc (Hons) Creative Digital Media

Keeran Jamil
Telephone: +44 (0)20 8331 8021
Room: QM330
e-mail: K.Jamil@greenwich.ac.uk

BSc (Hons) Computing
BSc (Hons) Business Information Technology
BSc (Hons) Business Information Systems
BSc (Hons) Business Computing
BSc (Hons) Computing with Digital Media
BSc (Hons) Computing with Games Development

Peter Smith
Telephone: +44 (0)20 8331 8516
Room: QM333
e-mail: P.A.Smith@greenwich.ac.uk

BSc (Hons) Computer Science
BSc/BEng (Hons) Software Engineering
BSc (Hons) Computer Systems and Networking
BSc (Hons) Computer Security and Forensics

Please see Moodle for the project supervisors and their respective tutorial sessions.

In addition to this handbook further guidance and support is provided via your project supervisor and second marker during tutorial sessions, as well as on Moodle.

2. Aims of the Project

The aims of the course are to:

- Provide the student with the opportunity to research, specify, design, implement and test a software product to an appropriate level of professional competence.
- Encourage the student to evaluate critically the work of others and relate it to their own work where appropriate.
- Develop a student's ability to create, plan, organise and work independently on an appropriate product, drawing on and extending ideas, skills and techniques encountered during the programme of study.
- Develop the student's ability to appraise critically, by means of a written report, the product and the process of its production as well as lessons learnt during the course of the project.
- Develop the student's ability to evaluate critically the work of others and relate it to their own work where appropriate.
- Develop the student's ability to critically appraise their own academic, creative and technical practice, by means of a written report, the product outcome and the process of its production.

3. Delivery of the Course

The course is essentially student centred with support provided by a supervisor and second marker during tutorial sessions in the first semester and one-to-one sessions in the second semester. The resources on Moodle, as well as the project website, are also valuable resources. We will aim to inform you about the project process, course delivery and assessment methodology, plus the project lifecycle and areas identified by you.

You will be assigned a supervisor – a member of the academic or research staff. They will advise and guide you throughout the course of the project.

Key Texts

The following will guide students through the research and project development stages.

ISBN Number	Author	Date	Title	Publisher
0-273-67809-4	Philip Weaver	2004	Success in your Project	Prentice Hall
0-13-021972-X	Dawson, C.W.	2000	Computing Projects, A Student's Guide	Prentice Hall
0-566-082772	Gash, S.	2000	Effective Literature Searching for students	Gower

0-566-084902	Sharp, J.A. Peters J. & Howard K.	2003	The Management of a Student Research Project (3/e)	Gower
0-419-14669-1	Turk, C, and J. Kirkman	1989	Effective Writing: improving scientific, technical and business communication, 2e	E & FN Spon
0-335-206867	Cryer, Pat	2000	The Research Student's Guide to Success	OUPress

4. The Role of the Supervisor

4.1. What is the role of the supervisor?

It is to advise, guide, criticise and encourage you through the project process, but it is not to direct the work unless the work contributes to a larger research project. The tasks the role encompasses are to:

1. Approve the project proposal when it meets the associated programme's criteria.
2. Agree on the supervision structure and the associated time allocation.
3. Discuss a general project strategy and approve the outline project plan.
4. Discuss possible avenues for literature search.
5. Discuss possible research methods, if applicable.
6. Discuss/advise on hardware and software and liaise with support staff, if applicable.
7. Offer advise re progress and warn of major problems.
8. Agree with the student and second marker a convenient time for the presentation.
9. Be the first marker and complete the development and final assessment forms

4.2. What you may expect, but is not in within the supervisor's role

1. Rewrite/finalise your project proposal.
2. Tell you what to do.
3. Edit/rewrite drafts, write the software or configure the hardware.
4. Provide detailed feedback when it is submitted late without an agreed extension.
5. Give detailed feedback on the summary, conclusions and evaluation.
6. Perform intermediate plagiarism detection.
7. Negotiate on your behalf for an extension. These are the same as for all pieces of assessed work and are dealt with via the School's policy for extensions and application for extenuating circumstances.

4.3. What happens to the supervision role over the holiday period?

In doing your project you must be able to take personal responsibility and minimal supervision is required. However, in the case of students re-sitting the project, the School of Computing & Mathematical Sciences will provide staff throughout the Summer period to provide 'immediate support'. You must agree with your supervisor when and how they may be contacted over the holiday period.

5. Assessment Criteria

There are two components in assessing a project:

- The product, which is demonstrated at a scheduled time, and uploaded by the project submission date (this may be as a Word, PDF or zip-file).
- The report, which is uploaded by the project submission date deadline date.

The marking scheme for these two components can be found in Appendix A.

5.1.Deliverables

The deliverables for the project are:

- A project pitch, together with a formal project proposal (Appendix B)
- Demonstration of initial software templates, together with interim report (Appendix B)
- Final demonstration of software and findings, together with upload of final report.

Students are expected to devote approximately 600 hours to a project. The final project report should be between 10,000 and 12,000 words and you must demonstrate an appropriate command of the rules of report layout, spelling, punctuation, grammar, syntax and due regard to the use of figures, diagrams and references. Where this does not take place the mark awarded will reflect this.

The purpose of the demonstration is to allow the supervisor and the second marker to see the product running, to check that you are its author, and to help them to visualize the product when they read through your description of it in the project report.

Consult the marking scheme (Appendix A) to confirm what you will be assessed on, for both the demonstration and final report.

5.2.Submission of project deliverables

The project pitch will be scheduled with your supervisor and second marker, and the project proposal should be uploaded to the e-supervision system.

The demonstration of your initial software templates will be scheduled with your supervisor and second marker, and the interim report should be uploaded to the e-supervision system.

The final demonstration will be scheduled with your supervisor and second marker. For the final project submission you need to upload a digital copy (pdf) of your project report (this should be accompanied with your coursework header sheet downloadable in the same way you have used for previous coursework). You also need to upload a zip-file containing the product you developed.

Deliverables and their respective dates can be found on the intranet.

5.3.Standardisation of assessment

Supervisors will assess and award marks for the work carried out by you. A second member of the academic staff will also assess and award marks for the end product and project report. The marks awarded by both examiners will be weighted to calculate an overall mark for the project.

If there is a significant difference between the two marks then it will be adjudicated by a project panel. The panel will also adjudicate projects, which are deemed to be at distinction level, and those, which are deemed to have failed. Additionally, the panel will moderate a representative sample of projects that do not fall into any of the above categories. The panel comprises of the Director of Quality, Heads of Schools, Programme Leaders and subject specialists.

5.4.Applying for Extenuating Circumstances

The procedures are as per the School policy for extenuating circumstances. All requests must be accompanied with full supporting evidence and will be reviewed by the School Panel shortly after receipt of the request and you will be informed of the outcome.

For an extension download the form from the CIS Intranet, complete and return the form together with supporting evidence to the School Office.

For Extenuating Circumstances you will need to apply online through the Portal, where you should also provide supporting evidence (for instance, scanned copies of certificates). Confidential evidence can also be provided in person to the School Office

Note: The maximum project extension that can be given is two weeks. Given below are examples of problems that are valid reasons for an extension:

- Illness – medical certificate required.
- Serious personal problems – we will take advice from Student Services if you do not wish to provide evidence to a member of the academic staff.

However the following are not acceptable as valid reasons for an extension

- Hard disk or other storage media becomes corrupted (be sensible and keep a recent backup)
- University of Greenwich computer lab is unavailable for a few hours.
- Minor illness such as a cold

Valid reasons for deferrals include (and must be supported by evidence provided over a long period of time):

- Illness over a long period of time
- Serious personal problems over a long period of time

6. Plagiarism

Plagiarism is presenting somebody else's work as your own work. It includes: copying information directly from the web or books without referencing the material; working with one or more other people on an individual coursework and submitting the joint coursework as your own individual effort; copying another student's coursework; paying someone else to do the work; stealing coursework from another student and submitting it as your own work. The person you copy from could be another student, a lecturer or someone outside the university. The School takes plagiarism very seriously and you have received a copy of the “Little Book of Plagiarism” to inform you how to avoid plagiarism.

- Plagiarism includes, but is not limited to:
- using published work without referencing (the most common)
- copying coursework essays
- collaborating with any other person when the work is supposed to be individual
- taking another person's computer file/program
- submitting another person's work as their own
- the use of unacknowledged material published on the web
- purchase of model assignments from whatever source
- copying another student's results
- falsifying results

All work suspected of being plagiarised is sent to the School's Academic Conduct Officer who investigates each case. Advice on how to reduce inadvertent plagiarism is available on the Student Intranet and Moodle, and gives advice on how to avoid being accused of unintentional plagiarism. Full details of the university guidelines can be found on http://www2.gre.ac.uk/current-students/regs.

Reuse of programming code

In industry reuse of code is to be encouraged and both Web sites and books will provide numerous examples of code BUT students should realise that part of the purpose of doing a programming coursework is to develop your skills. If most of the code comes from other sources then they will not be awarded a very high mark and you will have learnt very little.

If however you choose to make use of other people's code then in order to avoid an accusation of plagiarism, you must annotate their listing identifying the lines of code which are not your own. You must clearly state their source e.g. name of author, page in the book that they have taken the code from, Web page address. Failing to reference work taken from other sources is a plagiarism offence and will be dealt with as such.

Note you will be awarded more marks for the code you write yourself, than code used from others. Obviously if you copy the entire program from someone else (and reference the work) you will be awarded zero as you have not made a contribution to your project solution.

7. Regulations

7.1.Passing the course

To pass the course you must achieve an assessment mark of 40% or above.

You must complete your project by the deadline, unless an application for extenuating circumstances was approved. If you do not hand in your project or fail to meet the pass criteria for your submission deadline you will be referred to the re-sit deadline and the mark will be capped at 40%.

7.2. Reassessment regulations

If you fail the project module then you will be failed in the programme. If you do fail the programme you will have an opportunity to retake, re-attend and repay for the failed project course the following academic session.

8. Getting Started and Choosing a Project

The structure of the project will vary slightly, dependent on which area you study in, but you will be assigned a tutorial group and a set of co-supervisors at the start of the year. In the first semester you will be required to attend a weekly 1-hour lecture and 2-hour tutorial which will contain a set of activities that will help you start your project and enable you to get feedback on your project ideas. In Week 7, you will be required to submit a Project Proposal, which will be assessed and will account for 10% of your entire project mark. Shortly after that submission, there will also be a project pitch, where you will present your project idea to members of staff, and activities to help with areas such as academic writing, research, critical thinking, employability skills and project management.

In the second semester you will be allocated a first supervisor from one of your co-supervisors and you will then **be required to attend individual meetings** organised between yourself and your supervisor. Before submission of the final report, you will also be required to attend a project demo where you will be required to present your project deliverables and be asked questions by your supervisor and second marker (who will be one of your original co-supervisors).

8.1. Selecting a project topic

Your final year project is probably the largest individual piece of work that you will complete while studying with us for your undergraduate degree. It is vitally important that you choose a project which is appropriate to your skills and abilities, achievable within the limited time available, and of interest to you. A list of topics suggested by academic staff in your area of study are available, and you are strongly encouraged to select from among those. These topics are derived from staff research expertise and you are most likely to receive the best support if you choose from that list.

8.2. Choosing an appropriate project

Your project must reflect the programme of study that you are following. If it does not, then you will be in danger of getting a low mark, if not failing your project altogether. The weekly sessions will help to guide you in developing your ideas and interests into a suitable project proposal. Please note that, even when choosing from among the topics proposed by staff, the actual project you will carry out is defined by you in its detail, and agreed with the supervisor. A good project will begin by clearly identifying a question to be answered or a problem to be solved. Your project must contain an element of investigation or analysis of a problem. Your project must produce a product but development of a piece of software is not sufficient in itself. Your project will be mainly assessed by the report that you produce at the end your project. Your report will contain the following:

- A description of your project objectives
- An appraisal of relevant literature and data collected and identifying appropriate criteria for your project

- A description of the modelling/design/implementation of your solution
- An evaluation and reflection on the work that you have done

The detail of undertaking your project will be discussed in the project tutorials and you will get guidance from your supervisor, but these issues are worth considering this at this stage as you choose your project topic.

8.3.Finding a topic that is of interest to you

The important thing is that your project is something you feel comfortable about doing, that you have some interest in the topic and that you feel that you can achieve it. You should start with the list of topics that is available by your supervisors in your assigned tutorial group. These topics that can accommodate a wide variety of projects and you should approach your supervisors to discuss. On the other hand, it might be that you have already decided what your project topic is since before the year starts. That is fine and you can discuss this with your supervisors in September. However, if you are struggling to come up with an idea, then here is a simple check list to help you:

- Start by thinking about the things that interest you or you are associated with in some way. Ask yourself some or all of the following questions:
    - If you work (or you have done an industrial placement), are there any potential projects that you can find there?
    - Does a member of your family or a friend work somewhere that needs an application to be built?
    - What are your pastimes and hobbies? Do any of these suggest a possible project?
    - What are the areas you want to work on to get a job in your chosen career? What would impress a potential employer?
    - Is there something that you use every day that you think can be improved?
    - Are you a member of a club or association? Do they need an application to be built?
    - Can you find any ideas for a project topic by reading trade journals and magazines?
    - Can you find a topic that interests you from those suggested by members of staff?

  Use all sources available to you to try to come up with some ideas.

- Go through each of courses that you have studied so far. What topics interest you most from your programme? Can you find any ideas there? Identify a list of keywords for each of the courses you have studied.
- Using the material above, put aside some time to brainstorm (maybe sitting with a friend) and come up with as many ideas as possible. At this stage it does not matter how silly the idea is, but take twenty minutes to write down as many ideas as possible that come into your head. You should end up with a long list of possible projects.
- Go through your list and filter out the ideas which are impossible and try to whittle your list down to three potential projects.

- Using the list of keywords identified in (2), attribute these keywords to each of the project ideas you have selected. If you find that there are no keywords that seem appropriate, then it probably means that your project is not appropriate for your programme and you might not have the technical skills to complete the project.
- Next, spend a couple of days searching through material in books, on the internet, in academic papers, in newspapers and magazines, talking to friends and colleagues. (Try using the keywords you identified in earlier in a search engine and see what comes up.) Write down a short summary of what you would want to do for each of the three selected possible projects. You can then bring these along to the tutorials and discuss them with your supervisors.

8.4.Choosing an achievable project

One of the main causes of students failing their project is that they choose one that is either too simple or too complicated. Officially, the project lasts from the end of September to mid April (although it is useful to start doing background reading as soon as possible). So you have approximately 6 months, including the Winter and Easter break to complete your project. This is not a very long time and you should choose your project carefully, considering whether you can complete it within this timeframe.

Your supervisor will be able to give you advice as to whether or not the project is appropriate when you meet them at the beginning of your final year. For the time being, consider the following points:

- Do you have to learn any new technologies to complete this project?
- Are all the technologies you require for this project available?
- Do you have the right technical skills to complete this kind of project?
- Will you have to collect any data from external sources? If so, will it be available to you during the time you are completing your project?
- Are there any materials that you require which will be difficult to get hold of?
- How will you evaluate your product? How will you establish the degree of success in your project?

Use the answers to these questions to decide whether you are able to complete the project in the timeframe. The aims and objectives that you have considered for your project idea should also help in determining whether it is appropriate. If some of them require a considerable amount of work and you cannot guarantee that you can complete them in the timeframe, then it is probably too complicated. If you can complete all of the objectives within three weeks, then it is probably too simple.

9. Project Proposal (10% of the Project Mark)

This is the next stage of the project process following on from the project registration. Your Project Proposal is due in Week 7 of Term 1 and will account for 10% of your Final Year Project mark. Do some preliminary reading. Because most fields of engineering and science, and especially the field of computing, change rapidly, using textbooks alone is not enough; neither is concentrating solely on manufacturers' literature.

You could start with a trade journal, a scientifically or technically oriented general magazine such as Science, Scientific American, IEEE Computer or Byte, or ask a librarian to guide you to suitable publications in the topic area. The Internet is obviously a good place to start. The INSPEC database is held in the library on CD ROM and is a very good source of computing journal abstracts. Alternatively, to get a good grounding in a new topic area, find a good textbook. Decide first, what information to look for. The table of contents, preface and the introduction are a good general guide to the value a book is likely to have.

The following also need to be considered:

- If the project relies on 3rd parties (e.g. employers, clients, University personnel), can they be relied on?
- If a new skill needs to be acquired, how realistic is it that it can be mastered and what will happen if it can't?
- Can the project scope be changed if it turns out to be too simplistic or complex?
- How will the product be tested? If qualitative testing is used will there be sufficient users for a statistically relevant sample?

The answers to these questions are not easy and guidance may be needed – this will involve the support of your supervisor. The critical determinants of the success of a project are defining its breadth, depth and the supporting resources. The more effort that is done at this stage can often avoid problems later on.

Set measurable objectives. A clear objective is unambiguous – there is no mistaking what you intend to do. There should also be some very clear way of deciding when an objective has been met. The best – the only effective – objectives are those where it is possible to say whether or they have been achieved. By doing this you gain a measuring stick against which to judge everything that occurs during the project.

For each objective, there should be an associated statement that outlines how it will be achieved and how it will be measured once it is completed. For example, if one intended to evaluate types of hardware and then choose one, state how many types it is intended to evaluate it and how the criteria will set. This will produce a set of tasks and then it needs to be decided how manageable each of these tasks is, given one's circumstances and environment.

Use the resulting lists to determine what needs to be done in what order and estimate how long it will take e.g. visit a library, identify, obtain and read the relevant materials all need to be completed before it is possible to write up the literature survey. Similarly assembling the necessary software components must usually be completed before one can begin work on the system. Use the information to construct a project schedule.

This should show the schedule of tasks, associated descriptions and duration in hours per task. A Gantt chart should also be attached. Remember failure to plan, is planning to fail. It is much harder without a detailed plan – as many students will testify to.

You will also need to detail all of the resources required to successfully complete the project e.g. hardware software, access to data. It should also state whether or not there are any potential issues

that may arise, for example a company may be introducing new software next month but what are the implications if this is deferred for three months.

This should show the schedule of tasks, associated descriptions and duration in hours per task. A Gantt chart should also be attached. Remember failure to plan, is planning to fail. It is much harder without a detailed plan – as many students will testify to.

IMPORTANT: The template for your Project Proposal will vary depending on your degree subject area. Please refer to the electronic resources in Moodle, where you can find the relevant template under the links dedicated to your degree subject area.

10. Project Planning

10.1. Project milestones and deliverables

It is expected that your project plan will contain a set of milestones. Milestones are major points in your project work that you must pass in order to get to your goal. The University sets some milestones such as the dates for the submission of the project proposal, initial report, interim report, and the final report.

A deliverable should specify exactly what is to be delivered when some work has been completed. Deliverables include not just identifiable 'software', but also less tangible elements such as books and journals read, notes taken, tests performed, report drafts written. These should be tied closely to the milestones and hence to the objectives.

10.2. Timetable

Once the objectives, tasks, milestones and deliverables have been written down, they must be arranged in the form of a timetable. Estimate how many hours each task might require and over what period of time it will be completed. Set a date by which each milestone and deliverable will be achieved (remember some have been pre-set by the University and are non-negotiable). Review this critically to determine whether they are achievable given constraints of time and cost and then produce and maintain a Gantt chart.

11. Literature Review

11.1. What is the purpose of the literature review?

The literature review has three primary purposes:

".1 It justifies your project and shows that your project is not merely repeating the work of others, but has a contribution to make, perhaps by identifying a current gap in the literature of your field of study which you intent to fill.

.2 It sets your project within context by discussing and critically evaluating past and current research in your area. Through this contextualization you will identify how your project fits within, and contributes to, wider issues.

.3 It provides other researchers with a starting point from which they can understand how your project evolved and to identify what literature is relevant to your project in order that they can continue where you left off."

Dawson "*The Essence of Computing Projects*", 2000

It will also help the student to learn more about the topic area; define previous work in your proposed field and identify significant sources of information in the area you are studying. When working in isolation it is unlikely one would be aware of other potential solutions to the area under investigation. Therefore, it would not be possible to make any judgments about whether the solution is as effective as other solutions or not.

When completing a Literature Review there are four iterative steps:

- Define the search
- Perform the search
- Evaluate the material
- Write the literature review

The outcome will help to set the context to the problem area and allows the student to take an informed and critical viewpoint about the work.. This process also has the advantage of preventing the student from reinventing the wheel – use it to see if someone else has already carried out the work which is being proposed and, if so, is there some way it can be improved upon? They should also establish what kind of literature they are looking for before they start. Below are listed various source types:

- Books: these tend to be the first resort of students anyway and a good book will help the student refine their search and give them the necessary background information.
- Journal articles: these are especially important; they are recent and can give an view of the current state of the art.
- Conference literature: this can be quite good in providing a snapshot of a topic and can give you an idea of current research areas.
- The Internet: make sure you find trustworthy sources.

A copy of all articles studied must kept in case they are needed later in the project process and details of those read should be kept in the logbook with brief notes on each.

11.2. Writing up the results of a literature review

The review should provide "a coherent argument that leads to the description of a proposed study" (Rudestam and Newton, 1992) and should not just be a report that lists all the papers and books one has read with brief annotations on each one. It should convey to the reader what knowledge and ideas have been established on a topic, and what their strengths and weaknesses are. It will also demonstrate the student's ability of:

- information seeking: identifying a set of useful articles and books and scanning the literature efficiently

- critical appraisal: applying the principles of analysis to identify unbiased and valid studies

It should appear, at least in abbreviated form, as early as the initial report and most certainly as part of the interim report. It may be fully integrated into the main text of the final report or can appear as a separate chapter. There should be a full list of books and articles that have been consulted in the project bibliography.

12. Initial Report

The aim of this report is to inform the supervisor of the current project status and to review the feasibility of the project and objectives. It is important that this procedure concentrates on the key issues and that it requires the minimum of effort on the student's part. The report should give a summary of the overall project status. It should include a header page and the following sections:

Current Situation

In this section, describe the completed work. This should include an account of the literature survey to date – what books and articles have been consulted and how the material fits in with what one is planning to do.

Review of Objectives

Restate the objectives in your project proposals and discuss any changes.

Project Management

Discuss the project management plan, as stated in your proposal, with any changes.

Problems

The problem areas should describe the critical factors that will affect the meeting of target dates. Now is the time to bring to your supervisor's attention any problems that have arisen so far. This includes problems in obtaining equipment and software – and access to key people. This acts as an early warning system and focuses attention on the matters requiring urgent attention. If it is found that the project scope was too ambitious and some aspect of it needs to be revised then there is still time to have a chance of completing on time.

Development Method

Discuss the method or framework you'll be using in this project.

Literature Review

See Section 11.

Draft Chapter

Create a framework for the main sections of your report and write the preliminary content for these sections.

Key Work over the Next Period

In this section specify methods, tasks and activities plus the estimates of time to complete them. Indicate the actual progress and identify any deviation from the original proposal and plan.

Bibliography

See Appendix B for a template.

13. Final Report

The project report is the outcome of your research project. It is a scholarly document, which must be able to fit in just one volume (including all appendices). It incorporates:

- the results of your literature survey;
- a report on the way in which you have carried out your analysis, and the reasons for your choice of techniques;
- a design and implementation based on your analysis;
- a critical re-evaluation of the project itself and its outcomes.

Major writing tasks during this stage tend to be a "middle-out" approach. Once the outline structure has been decided, write the main body of the report first and add the supporting sections afterwards. The document does not have to contain all the elements listed in 14.1, and may be adapted to suit the needs of the project. The typical order of writing is "middle-out" and follows the sequence outlined below:

- Finalise outline structure
- Write main body
- Write conclusions, recommendations and summary
- Compile appendices and bibliography
- Write introduction
- Write abstract, preface and acknowledgements
- Prepare contents list
- Write critical appraisal

13.1. Logical structure of the project report

The following is the standard structure for the report. More advice is available on the project website.

Title Page

The Title Page contains the project title, your name, the date (month and year) of submission, your project supervisor and the title of the undergraduate programme.

Abstract

The Abstract, occupying less than half a page, is a short description of the intention of the project.

Preface

The Preface includes any relevant observations that do not belong in the project itself. It is here that the justification for the project meeting the programme requirements.

Acknowledgements

It is customary to acknowledge any substantial help, with either the project work or the report, from people and other informal sources.

Table of Contents (including appendices)

The report should be divided into chapters each of which may be divided into sections which may again be divided into subsections and so on. Each chapter and numbered section should **have a title, and the contents page should list the most significant of these.**

List of Tables

Optional if no tables have been used in the report

List of Figures

Optional if no figures have been used in the report

Body of Report

In the body of the report, each chapter should start on a new page. Chapter headings should appear more important than section headings. The following usually have one or more chapters devoted to them.

Introduction

The first chapter of the report is commonly an introduction, which gives an account of the work to be done and the context in which it is to be done, usually providing background information about when, and in what circumstances, the project was conceived. This should include a clear statement of aims and objectives and the methods used in carrying out the work involved in each stage of the project.

Literature Review

A summary evaluation and critical analysis of the literature read, relevant to the project topic, together with its possible application to the project.

The "middle" chapters

They would usually include the description of work done and the presentation and analysis of results.

Closing chapters

The closing chapters commonly include a summary and a conclusion together with any recommendations. In summarising, highlight the important stages and outcomes of the project. The conclusions, would normally consider and comment critically upon the results of the project; this includes both the process and the product. This should include a consideration of the extent to which the aims of the project have been achieved. Finally, recommend ways in which the work could be applied or extended.

List of References

References should be ordered alphabetically by the name of the author (or, if there is more than one, the name of the first author. The Harvard system is used (more information is available on the project website) and each reference should state the author's name and initials, date (in parentheses), title, publisher and place of issue (if known) e.g.

Seber G.A.F. (2003), Multivariate Observations, John Wiley, New York.

If the reference is to a journal or to a conference proceedings article, then the journal title, volume, number and page numbers should be added, e.g.

Parnas D.L. et al (2001), Evaluation of Safety Critical Software, CACM, Vol. 33, No.6, pp. 636-651.

Appropriate references should also be included if material has been used from lecture notes that have been written by members of staff, from previous students' project reports, or from laboratory manuals. These should cite the programme and course title together with the date of the course instance and the full name of the lecturer concerned.

References mentioned in the text of the report should be cited by the name of author and the date, inside parentheses, e.g.

"The Jackson technique (Jackson 2000) was employed for systems analysis and design after structured analysis (De Marco 1978) had been examined ….".

If you fail to properly cite sources of information you will lay yourself open to accusations of plagiarism.

Appendices

This should include detailed and technical documentation such as table of results, diagrams, program source code, etc, which are essential parts of the project but not directly a part of the main discussion in the report. All contents of appendices should be exclusively, products of the student's own work.

Other materials used during the project work (such as information from user manuals, interview notes, etc), which it is necessary to include, should if possible be summarized to only a few pages before entering into the appendix. Original copies of such material should be kept by the student and may be required to be produced as supporting evidence of their work. Examples of key coding may be provided in an Appendix but generally it should be on the P drive with its associated software.

13.2. Report layout

The following are guidelines for a standard report.

13.2.1. Size

The complete report should be capable of being bound in one volume. The normal size of a project report is on average 8,000 words – it should not exceed 12,000. The exact length of the report will depend on the type of project carried out.

Note: the number of words is counted from the beginning of the first chapter up to and including the end of the last chapter.

13.2.2. Page Layout

The report should be:

- one and a half line spacing.
- margins should be 2.5 CIS/1" all round.

13.2.3. Font size and type

It is recommended to use Times New Roman font type and font size 11.

13.2.4. Page numbering

Numbering from the Title Page through to but not including the first chapter should use small Roman numerals. The number on the Title Page (i) is not normally shown. The body of the report consisting of the first through to the last chapter, and all remaining report elements through to the very last page of the report, should be numbered using Arabic numerals (starting at 1).

13.2.5. Table of contents

Chapters should be numbered (1,2,…) and each section sub-numbered (2.1, 2.2, 2.3 …). Further decomposition into subsections should be numbered (2.2.1, 2.2.2, 2.2.3 and so on). Each chapter and numbered section or subsection should have a title, and the contents page should list the most significant of these.

13.3. Writing up

Written communication is extremely important in project work. Computer software is an intangible product, which can be difficult to assess as it progresses. For a long period of the development the reports may be all the student can show their supervisor. There are two reports to write for this course: an initial interim report (see section 13), and a final report (discussed in this section).

Appendix A: Project Marking Scheme

PROJECT MARKING SCHEME

Supervisor:

Second Marker:

Submission date:

DYSLEXIC STUDENT
Marks should not be deducted for presentation unless this is one of the learning outcomes being assessed.
Do not indicate spelling or grammatical errors unless the student specifically asks you to do so.

Broadly, the final Undergraduate project should be assessed using the following grade bands.

Marks	Undergraduate Criteria
>=80%	Meets all criteria. Shows a significant amount of critical analysis and exhibits an excellent understanding of the relevant issues. Product meets requirements.
70-79%	Meets most of the criteria. Demonstrates clear awareness of relevant issues with a high standard of critical analysis. Product meets requirements.
60–69%	Meets most of the criteria. Demonstrates clear awareness of relevant issues with a high standard of critical analysis. An attempt has been made at analysis using appropriate frameworks but may include some errors.
50-59%	Some of the criteria present but is mainly factual and descriptive with little grasp of analysis.
40-49%	Some of the criteria present and establishes a few relevant points but superficial or confused exposition of issues.
<40%	Little or no evidence of given criteria and no grasp of analysis. Does not demonstrate self-direction, originality in problem solving or a critical self-evaluation of the project process.

Overall assessment
NOTE: This report will be made available to students along with their marks after moderation.

Overall assessment of the report:

Overall assessment of the demonstration:
NOTE: This report will be made available to students along with their marks after moderation.

Have you held a demonstration or viva for this project?

Comments on the viva/demonstration

Does this project meet the requirements of the British Computer Society?

Dissertation projects are not accepted for BCS.

To meet the BCS criteria, the student needs to have:
(a) a mark of 40% or above
(b) produced a project which is practical and has problem solving in its content and relates to the development life cycle of computerised systems

Has the report met presentation criteria?

Comments on the Standard of Presentation (if not commented on above)

Reports should have a length of 8,000 - 12,000 words. They must also demonstrate an appropriate command of the rules of report layout, spelling, punctuation, grammar, syntax with due regard to the use of figures, diagrams and references. Where a candidate fails to satisfy assessors with respect to these rules the grade awarded should reflect this.
Note: All technical documentation should be provided as Appendices and can be excluded from the word count.

Authenticity of the Submission
All final reports are uploaded into TurnItIn for plagiarism checking.
Supervisors must inspect the TurnItIn report and comment on it.
Note that you must still report plagiarism using the CIS Assessment Offences System.

TurnItIn Percentage Match = %
Comments on the TurnItIn report, and any other plagiarism issues.

Each of the sections below nominally carries equal weight and the list of things to consider under each heading is indicative.
These may vary depending upon the project's aims and objectives.

---

1. **Understanding of the Problem Domain**:
   Demonstration:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   Report:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   This section is about assessing the student's ability to identify and investigate a suitable problem, and follow an appropriate project methodology to solve the problem.
   • Identified an area to research or investigate and a problem to be solved
   • Demonstrated understanding of the problem domain
   • Shown how project objectives were formed and project planning took place
   • Discussed the research or investigation within the context of the project
   • Critical evaluation (of the appropriateness) of the current thinking on the research area
   • Compared similar products and systems
   • Selected suitable criteria for development of product and ideas
   • Chosen appropriate tools for modelling and development
   • Production of requirements specification/client brief of suitable complexity
   • Identified any legal, social, ethical and professional issues that are relevant to the project

2. **Development of Product and Ideas**:
   Demonstration:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   Report:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   This section is about assessing the level of the development and the student's competency within the context of the chosen topic area.
   • Clear plan of how the hypothesis might be tested
   • In-depth investigation of the context/literature/other relevant products
   • A reasoned thread going throughout the project
   • Use of suitable data collection techniques
   • Use of suitable analysis tools to evaluate data
   • Demonstrated complexity in the design and implementation of the product
   • Discussed the development process

- Shown that a number of alternative approaches have been considered
- Explained the reasons for selecting a particular solution
- Demonstrated changes in the project plan
- Resolution of relevant legal, social, ethical or professional issues

3. **Product Build and Evaluation**:
   Demonstration:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   Report:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   This section is about assessing the student's ability to present their completed work and discuss issues of quality, usability, etc.
   - Demonstrated technical ability in building the product
   - Demonstrated the full scope of product developed
   - Shown that the product has been tested and evaluated appropriately
   - Discussed the quality of the product in relation to original objectives and criteria
   - Demonstrated the usability and appropriateness of the product for the problem domain
   - Identified where and how improvement can be made

4. **Conclusions and Critical Review**:
   Demonstration:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   Report:
   >=80%, 70-79%, 60–69%, 50-59%, 40-49% or <40%

   This section is about assessing the student's ability to be critical of their own work and show reflective thinking.
   - Demonstrated critical thinking in writing up the project
   - Discussed lessons learnt whilst completing the project
   - Identified any problems encountered and discussed how they were tackled
   - Identified mistakes made and lessons learnt
   - Reflected on how the project plan changed during the development
   - Made suggestions as to how the work can be improved
   - Identified how the project might be taken further or expanded

Appendix B: Initial Contextual Report Template

COMP1682 Final Year Project

Initial Contextual Report

Your Project Title

Your Name:

Banner ID:

Your Degree Title

Supervisor: Your Supervisor's Name

Due date: 15th December 2017

Word count:

Abstract

This is where you succinctly describe your project, the context of and interest in your project, your approach to your project and your progress to date using somewhere between 60 and 120 words. This is much easier to write after you have written the rest of this report.

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Introduction

Background

You have already written this in your proposal. This is an opportunity to improve on your original text.

Aims and objectives

You have already documented these in your proposal. This is your opportunity to state why you have changed your mind and provide revised objectives

Approach

Again, you have already discussed this in your proposal but you may wish to restate this now that you are older and wiser. The key here is to realistically describe your approach and avoid regurgitating tedious guff about DSDM or some other branded 'methodology' when your approach is truthfully just making it up as you go along and keeping your fingers crossed.

Justification of the suitability of a Methodology or a Framework followed.

Literature Review

Approach to literature searching

Discuss where you have looked and what you have looked for.

Identifying the problem

This will hopefully expand what you have looked for into what you have found

Another Sections

This is not a review of technology; it should be a review of literature. Do not simply summarise each source in sequence. Some published literature reviews are actually literature catalogues, for example Portocarrero et al. (2014). While these are useful documents this is not what we are looking for here. Your literature review should not be a compilation of facts and feelings but a coherent argument that leads into the description of the proposed study (Rudestam & Newton, 2007, p. 47). According to Rudestam & Newton (2007, p. 48) a common misunderstanding is that a literature review is a "*laundry list of previous studies with sentences or paragraphs beginning with words such as Smith found that… or Jones concluded that …*". Use a tool to manage and automatically generate your bibliography. I have used APA 6th edition to provide formatting that is remarkably similar to the Greenwich Harvard specification.

Yet Another Section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Conclusions

What are you taking forward from this? How is this going to influence what you are going to do next?

Product Research

4.1 Comparing of similar products or platforms against a set criteria for Usability purpose.

Legal, Social, Ethical and Professional Issues and Considerations

You may find it helpful to include a list of all applicable laws and perhaps other laws that you have considered and excluded, with reasons why they do not apply, if appropriate. It is generally not a good idea to decompose this section into sections labelled as 'Legal', 'Social', and so on as these aspects are intimately related.

Requirements

Analysis of requirements

In this project there are of….

Comparison of systems

For some projects there may be similar existing systems. This would be a good place to identify these, compare them and elicit requirements from them.

Requirements elicitation

Perhaps you have conducted some actual research with questionnaires and interviews and such like (having obtained research ethics clearance of course). This would be a good place to discuss this.

Functional requirements

A clear statement of functional requirements is the starting point for your design. Without this you will look foolish.

Non-functional requirements

Clear statement of non-functional requirements. It may be an idea here to refer back to your earlier LSEPi section as compliance is likely to (and should) be an important aspect here.

Design

Describe your approach to design. If perhaps you are using UML use-case notation then the line between requirements and design starts to blur (Šilingas & Butleris, 2015). Other UML notation is however part of the design process, but perhaps overlaps with testing.

Prototype

Describe your approach to creating your first (and second and third) prototype(s).

Review of technology

If you need to review technologies this is perhaps the place to do it. Perhaps you need a data base, in which case state what database engines have you considered, which have you decided to use and for what reason. Perhaps you need an app, in which case you could, for example, state which development frameworks are available, compare and contrast all, select 3 for detailed comparison, choose one and state why you have made this choice.

Reflection

Wrap the whole thing up with some conclusions. A reflective approach is increasingly popular but many students struggle with introducing a critical personal perspective (Day, 2013).

Bibliography

Day, T. (2013). *Success in Academic Writing*. Palgrave.

Portocarrero, J. M., Delicato, F. C., Pires, P. F., Gámez, N., Fuentes, L., Ludovino, D., & Ferreira, P. (2014). Autonomic Wireless Sensor Networks: A Systematic Literature Review. *Journal of Sensors*.

Rudestam, K. E., & Newton, R. R. (2007). *Surviving Your Dissertation: A Comprehensive Guide to Content and Process* (3rd ed.). SAGE.

Šilingas, D., & Butleris, R. (2015). Towards implementing a framework for modeling software requirements in MagicDraw UML. *Information Technology and Control, 38*(2).

Appendices